

Custom Arcade Cabinet

Visual Inspiration



Project Description

When I first arrived at Brown, there was a student made arcade cabinet in the basement of the student center. I was always a huge fan of retro arcade games when I was younger - particularly Centipede, Dig Dug, and Marvel vs. Capcom - so, I was really fond of this student creation. However, within six months, the cabinet's screen stopped working and it was thrown away. I always thought that it would be a really cool project to make a new arcade cabinet and document my process so that it could continue to be a fun activity for students for many years to come. I think this would be a good way to apply my skills in CAD, laser cutting, and design. I also think it would be a great opportunity to learn more about soldering, CNC woodworking, wood finishing, and latex printing for decals.

For my final product, I hope to have an arcade cabinet utilizing Raspberry Pi to run emulators on RetroPi. My cabinet will have 4 player capabilities, a spatial sound system, a high refresh rate display, wheels for easy transportation, and a LED marquee.

Build Plan

My timeline to build my project would be over the course of approximately 4 months, from September to December, until winter break starts. I want to be sure to take my time so that the product is the best it could possibly be. I would need to learn some new

skills, like advanced wood working, the wood CNC machine, and perhaps some soldering. My time line would be approximately as follows:

Month 1 (September)

- Take advanced woodworking and CNC workshops as necessary
- Perhaps do laser cut test construction for mini cabinet model
- Complete ordering of materials
- Design layout of 4 player buttons

Month 1 (October)

- After wood received: pre-treat wood, cut cabinet pieces (<https://www.classicarcadecabinets.com/assembly---plywood.html>)
- Begin testing of retropi on raspberry pi (<https://www.lakka.tv/disclaimer/>)
- Do soldering for speakers, assembly of other electronics for testing

Month 1 (November)

- Laser cut acrylic for marquee, screen protector for monitor
- Do cabinet assembly
- Do full electronic assembly, integrate into cabinet (<https://www.instructables.com/How-to-Make-an-Arcade-Machine-for-Cheap/>)

Month 1 (December)

- Finishing touches, vinyl cut decals for final cabinet
- Perhaps integrate Nintendo switch dock, any other additional features

Materials/Budget

I anticipate the project costing approximately \$501. One place where I could save money within my budget is perhaps the sourcing of plywood, which I currently list from Home Depot. I would also hope to have access to materials like paint or wood treatment to add a nice finish to the cabinet. Perhaps if there are extra pre-used raspberry pi's available around campus, I could utilize those instead of ordering a brand new one. I would be very open to any other sourcing suggestions where I could save money in my budget.

https://docs.google.com/spreadsheets/d/1s4Nf_eH2LhgOTly47memETmgfMb_fDe-p-355-4vtX4/edit?qid=0#qid=0

System category	Item	Supplier	Web link	Relevant specs	#	Price
Display	Monitor	Amazon	https://www.amazon.com/Viewedge-Monitor-Con	27", 144 Hz		105.99
Display	Marquee	Amazon	https://www.amazon.com/dp/B088BTXHRG/ref=			16.99
Game System	Buttons	Amazon	https://www.amazon.com/dp/B071WC7TGK/ref=	illuminated, raspberry pi compatible		83.99
Game System	Raspberry Pi	Amazon	https://www.amazon.com/CanaKit-Raspberry-Sta	8 GB, 4 USB ports, 2 HDMI cables, active cooling		159.99
Game System	Speaker	Amazon	https://www.amazon.com/dp/B07GP9MLS8?ref=			14.98
Structure	Plywood	Home Depot	https://www.homedepot.com/pep/Columbia-Fore	3/4 in. x 4 ft. x 8 ft.		89.88
Structure	Caster Wheels	Amazon	https://www.amazon.com/ASHGOOB-Casters-Loc	4 Pack 2"		14.98
Structure	Surge Protector	Amazon	https://www.amazon.com/Extension-Multiple-Out	10', 6 outlets		14.98
X-Alternative	Speaker	Amazon	https://www.amazon.com/AmazonBasics-USB-Pc	USB and audio ports		18.37
X-Alternative	Monitor	Amazon	https://www.amazon.com/dp/B0F9QM13WQ?tag	27" Display		139.99
X-Alternative	Marquee	Amazon	https://www.amazon.com/EDIY-1280x400-Raspb	6.9" display		39.79
X-Optional	Switch Dock	Amazon	https://www.amazon.com/Docking-Station-Ninter	NA		8.48
				Total (necessary)		501.78

Additional Resources

BDW Links

<https://www.browndesignworkshop.org/grants>

https://docs.google.com/forms/d/e/1FAIpQLSfGQ2L-sJ_yraaOPq6lTPiVlbnLRmqCO2LZE7Z98oqEe3Wolw/formResponse

How To Build Links

Description of CNC Plywood + Paint Process

<https://www.classicarcadecabinets.com/assembly---plywood.html>

Potential Arcade Layouts

<https://www.classicarcadecabinets.com/mortal-kombat-2.html>

Construction + General Process

<https://www.instructables.com/How-to-Make-an-Arcade-Machine-for-Cheap/>

<https://retromash.com/2016/02/09/building-a-home-arcade-machine-part-3-cabinet-design/>

Framework

<https://www.classicarcadecabinets.com/teenage-mutant-ninja-turtles-4-player.html>

Emulators for Raspberry Pi

<https://www.lakka.tv/disclaimer/>

Switch Docks

https://www.amazon.com/Dnkeaur-Nintendo-Docking-Accessories-Portable/dp/B0C89CWP15/ref=sr_1_3?crid=3T5H25QKZQSD9&dib=eyJ2ljojMSJ9_hifBoNmXHRXnK3Fav9HiLq1zZQsIGVbmEDlyS0jiuZFCg2sjm8PwO4NZE9rcf4E3Jt8hxjuM2zoKNSAZF5veDHTMu0oDhxFgXFPN7HNchNdQtxu07gD3hNaq8EHmYchXc_86xxWa-k3iA10hNbVPf1BwKEM7xFm2nEyIWZmAXu1EugoGe0B97y4lv53YrRIS86MhQkRuZQu-xq-l_CAPgRAoQXFKPjy6uiqyX5Ctas.P2A02RFbM5ODYvMbJfECaX9Q8ERR1P5mykK24m5Mruw&dib_tag=se&keywords=Switch%2Bdock&qid=1737994854&sprefix=switch%2Bdock%2Caps%2C258&sr=8-3&th=1&psc=1

Arcade Buttons

https://www.amazon.com/dp/B071WC7TGK/ref=sspa_dk_hqp_detail_aax_0?psc=1&sp_csd=d2lkZ2V0TmFtZT1zcF9ocXBfc2hhcmVh

https://www.amazon.com/gp/aw/d/B0CN6KGB2W/?encoding=UTF8&pd_rd_plhdr=t&aaxitk=1f35f24d91aa495c48e36640271644a7&hsa_cr_id=0&qid=1749108790&sr=1-2-3c6b3b04-89d4-46ee-857c-1e2f0de6a70e&ref_=sbx_be_s_sparkle_dlcasin_1_rating&pd_rd_w=rZ6Q4&content-id=amzn1.sym.4870a952-0dfa-4beb-9d2a-7a52537f019d%3Aamzn1.sym.4870a952-0dfa-4beb-9d2a-7a52537f019d&pf_rd_p=4870

[a952-0dfa-4beb-9d2a-7a52537f019d&pf_rd_r=25W168D6366SA50XDF9D&pd_rd_wg=q2j0Z&pd_rd_r=176e7dc-f461-42b7-a39b-499570cc9c1e#customerReviews](https://www.amazon.com/dp/B0816XSLM7?ref=asc_df_B0816XSLM7?mcid=19ef5e2f8a7c393eaea40b58a44dab5b&hvociid=11693360878234757503-B0816XSLM7-&hvexpln=73&tag=hyprod-20&linkCode=df0&hvadid=721245378154&hvpos=&hvnetw=g&hvrand=11693360878234757503&hvpone=&hvptwo=&hvqmt=&hvdev=m&hvdvcmdl=&hvlocint=&hvlocphy=9011867&hvtargid=pla-2281435177658&psc=1)

<https://www.diyretroarcade.com/collections/arcade-parts-buttons>

<https://www.amazon.com/stores/Sehawei/page/E8B38287-CDFE-4D9E-9804-9C0B43319D5E/feed>

https://www.amazon.com/SJ-JX-Controller-Microswitch-Raspberry/dp/B0816XSLM7?ref=asc_df_B0816XSLM7?mcid=19ef5e2f8a7c393eaea40b58a44dab5b&hvociid=11693360878234757503-B0816XSLM7-&hvexpln=73&tag=hyprod-20&linkCode=df0&hvadid=721245378154&hvpos=&hvnetw=g&hvrand=11693360878234757503&hvpone=&hvptwo=&hvqmt=&hvdev=m&hvdvcmdl=&hvlocint=&hvlocphy=9011867&hvtargid=pla-2281435177658&psc=1

Monitors

<https://www.amazon.com/dp/B0F9QM13WQ?tag=track-ect2-usa-73394-20&linkCode=osi&th=1&psc=1>

https://www.amazon.com/VSDISPLAY-2880x864-Stretched-Monitor-Computer/dp/B0D5Y65DL2?ref=asc_df_B0D5Y65DL2?mcid=07fa538bb6a63afe91821ae9c5ae6cf8&hvociid=12264911616949667104-B0D5Y65DL2-&hvexpln=73&tag=hyprod-20&linkCode=df0&hvadid=721245378154&hvpos=&hvnetw=g&hvrand=12264911616949667104&hvpone=&hvptwo=&hvqmt=&hvdev=m&hvdvcmdl=&hvlocint=&hvlocphy=9011867&hvtargid=pla-2281435177618&psc=1

https://www.amazon.com/insignia-fire-tv-32-inch-class-f20-series-hd-smart-tv/dp/B09ZLTMWWH/ref=mp_s_a_1_4?adgrpid=176014917066&dib=eyJ2ljojMSJ9.Gxk-LmvzI2JGCM9e4vjspjmZAljbFtOKc4rtNMGSDn36z7UgDX3ITVYBWwPqcmZDfQLN08tvJDwff-g6ChqPzonfktXbZVru2lluyLu-ueE9ozgDb-YCm-Y24ifOb0Td8Chrv1h2zQ9JXhjyhiEBU8fXZv56Rxp2ISxF27BUqDVY0aeOiT45R8INEpaBVBJu88hL1rapFw2fkgIz004wfg.EZxkiQixCsLZ3429sUmw1CTMLqt4yOMCUJzLcM9430U&dib_tag=se&hvadid=739151677090&hvdev=m&hvexpln=68&hvlocphy=9052452&hvnetw=g&hvociid=9428082663732543327--&hvqmt=e&hvrand=9428082663732543327&hvtargid=kwd-11147511&hydadcr=26738_11676259&keywords=television&mcid=37375b0696923602b870da739c26dfda&qid=1749132658&sr=8-4

[https://www.top10-usa.com/Budget%20Monitors?targetid=kwd-330612241864&matchtype=e&device=m&campaignid=22426426729&creative=745000132514&adgroupid=178505490220&feeditemid=&loc_physical_ms=9052452&loc_interest_ms=&network=g&devicemodel=&placement=&keyword=\\$best%20budget%20monitors&target=&aceid=&adposition=&trackid=us_top10_new_domain_1_1&mld=407-132-4411&qad_source=1&qad_campaignid=22426426729&qbraid=0AAAAACWZmZlaG-2tnT9dYE_KnGbtVcQCa&qclid=Cj0KCQjwglXCBhDBARIsAELC9ZiU-Zh1V09uaCAEbQH2r2n3NH2Dxvxn3BQDbhleBGba9_eWKnaUva4aAintEALw_wcB](https://www.top10-usa.com/Budget%20Monitors?targetid=kwd-330612241864&matchtype=e&device=m&campaignid=22426426729&creative=745000132514&adgroupid=178505490220&feeditemid=&loc_physical_ms=9052452&loc_interest_ms=&network=g&devicemodel=&placement=&keyword=$best%20budget%20monitors&target=&aceid=&adposition=&trackid=us_top10_new_domain_1_1&mld=407-132-4411&qad_source=1&qad_campaignid=22426426729&qbraid=0AAAAACWZmZlaG-2tnT9dYE_KnGbtVcQCa&qclid=Cj0KCQjwglXCBhDBARIsAELC9ZiU-Zh1V09uaCAEbQH2r2n3NH2Dxvxn3BQDbhleBGba9_eWKnaUva4aAintEALw_wcB)

<https://www.amazon.com/dp/B0D1DZNL3H?tag=track-ect2-usa-73394-20&linkCode=osi&th=1&psc=1>

Marquees

<https://www.amazon.com/dp/B0CFY477Q7?tag=avsklmax-20>

https://www.amazon.com/EDIY-1280x400-Raspberry-Monitor-Vehicle/dp/B0C3YFHF MJ/ref=mp_s_a_1_17?crd=7W22UEWFUBDG&dib=eyJ2ljojMSJ9.w0H57-CQlqG2LiHpXSdQV36ldtodks4Amo6Cn2Sk9VD8hqqNd5w61xf_ScVsP8I0OI2Ygxbj2INxXrz_vztPS5Phs1d5spuSAxoqUb-tnmPqjFMVV3Fg3bAT4RdEWxD7C1OBZmTYyre_ZKOc_fKRfMD1pfTZM6as9r10mk_MtOk-qOW7LIK5lwhczw106BhCR5XRqkfXA8HZKjQA

[ASW3GA.dYJeAKFvY40duvKk8KxLQArlyeN4edn-owuf6_cS2il&dib_tag=se&keywords=8+bit+long+display&qid=1749132930&srefix=8+bit+long+display%2Caps%2C93&sr=8-17&xpid=VPSU_df0-RI1z](https://www.amazon.com/dp/B0F6NPCX1V/ref=sspa_dk_detail_6?psc=1&pf_rd_p=f2f1cf8f-cab4-44dc-82ba-0ca811fb90cc&pf_rd_r=1KNJYF8BQQFYNDY7F56&pd_rd_wg=XXwL4&pd_rd_w=3iYe4&content-id=amzn1.sym.f2f1cf8f-cab4-44dc-82ba-0ca811fb90cc&pd_rd_r=5b007eee-4424-4a0c-bf2f-64215c36dbe5&sp_csd=d2lkZ2V0TmFtZT1zcF9kZXRhaWxhdGhlfWF0aWM)

https://www.amazon.com/dp/B0F6NPCX1V/ref=sspa_dk_detail_6?psc=1&pf_rd_p=f2f1cf8f-cab4-44dc-82ba-0ca811fb90cc&pf_rd_r=1KNJYF8BQQFYNDY7F56&pd_rd_wg=XXwL4&pd_rd_w=3iYe4&content-id=amzn1.sym.f2f1cf8f-cab4-44dc-82ba-0ca811fb90cc&pd_rd_r=5b007eee-4424-4a0c-bf2f-64215c36dbe5&sp_csd=d2lkZ2V0TmFtZT1zcF9kZXRhaWxhdGhlfWF0aWM

Plywood

https://www.homedepot.com/pep/Columbia-Forest-Products-3-4-in-x-4-ft-x-8-ft-PureBond-Birch-Plywood-165921/100077837?source=shoppingads&locale=en-US&pla&mtc=SHOPPING-BF-CDP-GGL-D21-021_001_PLYWOOD-NA-NA-NA-PLALIA-NA-NA-NA-NA-NBR-NA-NA-NEW- PLATEST&cm_mmc=SHOPPING-BF-CDP-GGL-D21-021_001_PLYWOOD-NA-NA-NA-PLALIA-NA-NA-NA-NA-NBR-NA-NA-NEW- PLATEST-22114087436-173523647899-299067763894&gad_source=1&gad_campaignid=22114087436&gbr aid=0AAAAADq61UeN36n7P6CUC8WS1Zpc7oDtF&gclid=CjwKCAjw3rnCBhBxEiwArN0QE5gmTEtH6m8jufGgfRBAZOuwVvtKmNOda2QEA S4u9a2KHZ5Ng190BoCFrsQAvD BwE&gclsrc=aw.ds#overlay

Raspberry Pi

https://www.amazon.com/CanaKit-Raspberry-Starter-Kit-PRO/dp/B0CRSNCJ6Y?ref_=ast_sto_dp&th=1
<https://retroPie.org.uk/docs/>
https://www.canakit.com/raspberry-pi-retro-gaming-kit.html?srsId=AfmBOor_0qclXYiTfxtDhCZm29CcgzSajcKk0hbC2G4UM-_sAp5-7jL-

Vinyl Cutter

[https://www.google.com/search?q=Vinyl+Cutter+\(Silhouette+Cameo\)&ie=UTF-8&oe=UTF-8&hl=en-us&client=safari&safe=active](https://www.google.com/search?q=Vinyl+Cutter+(Silhouette+Cameo)&ie=UTF-8&oe=UTF-8&hl=en-us&client=safari&safe=active)

Arcade Cabinet Research

<https://www.youtube.com/watch?v=AAdkkuuNWJY>